

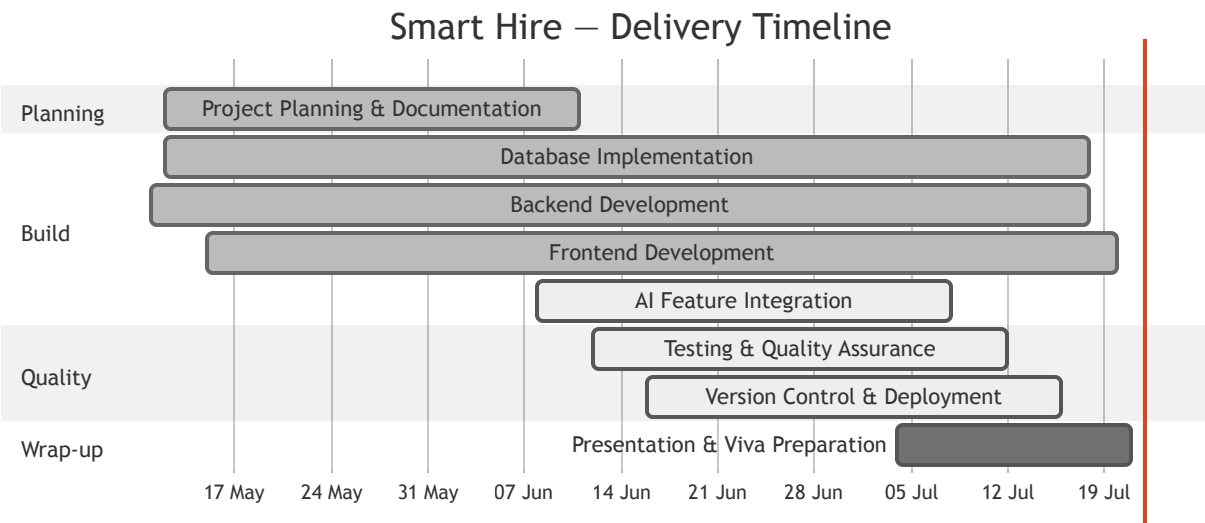
C-CLARKE INTERNATIONAL INSTITUTE OF DIGITAL SCIENCES

Work Breakdown Structure

Full task breakdown by epic, matching the project's Jira board.

Program	Diploma in Advanced AI & Software Engineering
Batch	25/26
Project	Smart Hire — Service Marketplace
Team	Nilan Iddawela, Janaka Eranda, Dasun Anjana, Damith Lasantha, Matheesha Milan
Date	21 July 2026

The Work Breakdown Structure below mirrors the project's Jira board (project CCSHP) exactly: eight epics, each broken into the individual tasks that were planned, assigned, and tracked to completion.



1. Full Breakdown by Epic

1. Project Planning & Documentation (CCSHP-4 · 100% complete)

1.1 SRS (Software Requirements Specification) DONE

1.2 Use Case Diagrams DONE

1.3 User Stories DONE

1.4 Work Breakdown Structure (WBS) DONE

1.5 Database Design DONE

1.6 API Documentation DONE

1.7 Testing Documentation DONE

2. Frontend Development (CCSHP-5 · 95% complete)

2.1 Design User Registration Page DONE

2.2 Design Login Page DONE

2.3 Develop Service Provider Dashboard DONE

2.4 Develop Admin Dashboard DONE

2.5 Develop Customer Profile Page DONE

2.6 Develop Provider Profile Page DONE

2.7 Develop Service Listing Page DONE

2.8 Develop Service Details Page DONE

2.9 Implement Service Category Filters DONE

2.10 Develop Service Booking Interface DONE

- 2.11 Develop Booking History Page DONE
- 2.12 Develop Review and Rating Interface DONE
- 2.13 Develop AI Recommendation Interface DONE
- 2.14 Integrate Authentication APIs DONE
- 2.15 Add Form Validation and Error Messages DONE
- 2.16 Implement Responsive Design TO DO
- 2.17 Test Frontend User Interfaces DONE
- 2.18 Fix Frontend Bugs DONE
- 2.19 Review and Finalize Frontend DONE

3. Backend Development (CCSHP-6 · 100% complete)

- 3.1 Integrate Service Management APIs DONE
- 3.2 Integrate Booking Management APIs DONE
- 3.3 Integrate Review and Rating APIs DONE
- 3.4 Implement Service Search DONE
- 3.5 Create API Endpoints for Reviews, Admin, Payments, and Health Check DONE
- 3.6 AI Assistance Phase 2 DONE

4. Database Implementation (CCSHP-7 · 100% complete)

- 4.1 Set Up PostgreSQL Database DONE
- 4.2 Create Users Table DONE
- 4.3 Create Service Providers Table DONE
- 4.4 Create Service Categories Table DONE
- 4.5 Create Services Table DONE
- 4.6 Create Bookings Table DONE
- 4.7 Create Reviews and Ratings Table DONE
- 4.8 Create AI Recommendations / Conversation Tables DONE
- 4.9 Define Primary and Foreign Keys DONE
- 4.10 Create Table Relationships DONE
- 4.11 Apply Database Normalization DONE
- 4.12 Create Database Constraints DONE
- 4.13 Create Database Indexes DONE
- 4.14 Create Database Migration Scripts DONE
- 4.15 Connect Backend to Database DONE
- 4.16 Add Sample Data DONE
- 4.17 Test CRUD Operations DONE
- 4.18 Test Database Relationships DONE
- 4.19 Test Data Integrity and Constraints DONE

- 4.20 Create Database Backup DONE
- 4.21 Fix Database Issues DONE
- 4.22 Review and Finalize Database Implementation DONE

5. AI Feature Integration (CCSHP-8 · 95% complete)

- 5.1 Select AI Feature and Define Requirements DONE
- 5.2 Design AI Service Recommendation Workflow DONE
- 5.3 Set Up OpenAI API DONE
- 5.4 Configure AI Environment Variables DONE
- 5.5 Prepare Service Data for AI Processing DONE
- 5.6 Develop AI Service Recommendation Logic DONE
- 5.7 Create AI Recommendation API Endpoint DONE
- 5.8 Connect AI Feature to Backend DONE
- 5.9 Develop AI Recommendation User Interface DONE
- 5.10 Connect AI Interface to Recommendation API DONE
- 5.11 Add AI Input Validation DONE
- 5.12 Add AI Error Handling and Fallback Responses DONE
- 5.13 Test AI Recommendations DONE
- 5.14 Improve Recommendation Accuracy TO DO
- 5.15 Test AI API Integration DONE
- 5.16 Secure AI API Keys DONE
- 5.17 Record AI Testing Evidence DONE
- 5.18 Fix AI Integration Issues DONE
- 5.19 Document AI Feature Implementation DONE
- 5.20 Review and Finalize AI Feature DONE

6. Testing & Quality Assurance (CCSHP-9 · 97% complete)

- 6.1 Prepare Testing Plan DONE
- 6.2 Define Test Scenarios DONE
- 6.3 Prepare Functional Test Cases DONE
- 6.4 Prepare Non-Functional Test Cases DONE
- 6.5 Test User Registration and Login DONE
- 6.6 Test Authentication and Authorization DONE
- 6.7 Test Customer Dashboard DONE
- 6.8 Test Service Provider Dashboard DONE
- 6.9 Test Admin Dashboard DONE
- 6.10 Test Service Creation and Management DONE
- 6.11 Test Service Search and Filtering DONE

- 6.12 Test Booking Process DONE
- 6.13 Test Review and Rating Feature DONE
- 6.14 Test AI Recommendation Feature DONE
- 6.15 Perform Backend API Testing DONE
- 6.16 Perform Database Testing DONE
- 6.17 Perform Frontend UI Testing DONE
- 6.18 Perform Integration Testing DONE
- 6.19 Perform Responsive Design Testing TO DO
- 6.20 Perform Security and Input Validation Testing DONE
- 6.21 Perform User Acceptance Testing DONE
- 6.22 Record Testing Evidence DONE
- 6.23 Create Bug Reports DONE
- 6.24 Fix Identified Bugs DONE
- 6.25 Retest Fixed Issues DONE
- 6.26 Perform Regression Testing DONE
- 6.27 Prepare Final Test Summary DONE
- 6.28 Review and Finalize Quality Assurance DONE
- 6.29 8 real defects logged and resolved (see Testing Documentation) DONE

7. Version Control & Deployment (CCSHP-10 · 64% complete)

- 7.1 Create Git Repository DONE
- 7.2 Configure Git Repository DONE
- 7.3 Add .gitignore File DONE
- 7.4 Create Development Branch DONE
- 7.5 Create Feature Branches DONE
- 7.6 Add Project Source Code to Repository DONE
- 7.7 Commit Frontend Source Code DONE
- 7.8 Commit Backend Source Code DONE
- 7.9 Commit Database Scripts DONE
- 7.10 Write Clear Commit Messages DONE
- 7.11 Create and Review Pull Requests DONE
- 7.12 Merge Feature Branches DONE
- 7.13 Resolve Merge Conflicts DONE
- 7.14 Tag the Final Project Version TO DO
- 7.15 Configure Environment Variables DONE
- 7.16 Prepare Production Database TO DO
- 7.17 Run Database Migrations DONE
- 7.18 Deploy Backend Application TO DO

- 7.19 Deploy Frontend Application TO DO
- 7.20 Connect Deployed Frontend and Backend TO DO
- 7.21 Test the Deployed Application TO DO
- 7.22 Fix Deployment Issues TO DO
- 7.23 Configure Security Settings TO DO
- 7.24 Verify Production System TO DO
- 7.25 Finalize Git Repository DONE

8. Presentation & Viva Preparation (CCSHP-11 · 0% complete)

- 8.1 Plan Presentation Structure TO DO
- 8.2 Prepare Project Introduction TO DO
- 8.3 Prepare Problem Statement and Proposed Solution TO DO
- 8.4 Prepare System Architecture Slides TO DO
- 8.5 Prepare Technology Stack Slides TO DO
- 8.6 Prepare Use Case and User Story Slides TO DO
- 8.7 Prepare Database Design Slides TO DO
- 8.8 Prepare Frontend and Backend Slides TO DO
- 8.9 Prepare AI Feature Explanation TO DO
- 8.10 Prepare Testing and QA Slides TO DO
- 8.11 Prepare Git and Project Management Evidence TO DO
- 8.12 Prepare Deployment Explanation TO DO
- 8.13 Add System Screenshots to Presentation TO DO
- 8.14 Prepare Live System Demonstration TO DO
- 8.15 Assign Presentation Sections to Team Members TO DO
- 8.16 Prepare Viva Questions and Answers TO DO
- 8.17 Review Complete Source Code TO DO
- 8.18 Review Database Design and Relationships TO DO
- 8.19 Review API Endpoints TO DO
- 8.20 Review Authentication and Authorization TO DO
- 8.21 Review AI Integration TO DO
- 8.22 Review Testing Procedures TO DO
- 8.23 Review Deployment Process TO DO
- 8.24 Conduct Practice Presentation TO DO
- 8.25 Conduct Mock Viva TO DO
- 8.26 Record Review Meetings TO DO
- 8.27 Fix Presentation Issues TO DO
- 8.28 Finalize Presentation and Demonstration TO DO

